Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your latest score.

Next item →

1. A team manually runs several batch jobs in a sequence. The process is slow and error-prone. What is the primary benefit of using a workflow orchestration tool? — 1 / 1 point

- ( ) It automatically scales compute resources to make jobs run faster.
- ( ) It automatically improves data quality within each batch job.
- ( ) It automatically encrypts data as it moves between jobs.
- (•) It automates task dependencies, scheduling, retries, and error handling.

> **Correct**
> Correct! An orchestrator automates the workflow, ensuring jobs run in the correct order, on schedule, and that failures are handled without manual intervention.

2. When designing a data pipeline, what does the principle of "idempotency" mean? — 1 / 1 point

- ( ) The pipeline can only be run one time and must be rebuilt to run again.
- (•) The task can be run multiple times with the same input yet produce the same final result.
- ( ) The pipeline is designed to run very quickly, regardless of data volume.
- ( ) The pipeline automatically updates its own logic from a source code repository.

> **Correct**
> Correct! For example, a job that overwrites its output table will produce the same result even if re-run. This is crucial for building safe, retry-able workflows.
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3. When triggering a cloud service from an orchestrator, what is the most reliable approach? — 1 / 1 point

- (•) Use a dedicated operator designed for that specific cloud service.
- ( ) Write custom functions that use client libraries to call the service.
- ( ) Use a generic operator to execute command-line scripts.
- ( ) Use a generic HTTP operator to call the service's public API endpoint.

> **Correct**
> Correct! This is the best practice, as it provides native integration for security, parameter passing, and job status monitoring, making the workflow more reliable and secure.

4. To make a workflow reusable across "development," "staging," and "production" environments, where should configuration details like project IDs or passwords be stored? — 1 / 1 point

- ( ) In a text file stored alongside the main workflow file.
- ( ) Hard-coded directly in the workflow definition file.
- ( ) In a database table that the workflow must query every time it runs.
- (•) In a centralized configuration system built into the orchestrator.

> **Correct**
> Correct! The workflow logic should remain generic, while environment-specific values are pulled from a managed configuration system. This makes the logic portable.